

MOHAMMED TALHA HUSSAIN

 +91 7993362331  imtalhahussain10@gmail.com  imtalhahussain  talha-hussain  Hyderabad, IN

SUMMARY

I am an AI Engineer specializing in **Generative AI, Retrieval- Augmented Generation RAG , and LLM-powered applications**, with experience building **end-to-end AI systems from data ingestion to deployment**. Skilled in developing **scalable backend architectures, real-time AI pipelines, and multi-agent systems** using FastAPI, vector databases, and modern LLM frameworks. Experienced in **data processing and pipeline development using SQL-based workflows and platforms like Snowflake and Databricks (working knowledge)**. Strong focus on **low-latency systems, production-ready AI solutions, and real-world deployment**.

EXPERIENCE

AI Engineer (Consulting – RAG & LLM Systems)

Independent Consulting

05/2024 - Present  Hyderabad, IN

Consulting services focused on AI and machine learning

- Designed and deployed end-to-end AI pipelines (ingestion → retrieval → generation API serving) for real-world applications using FastAPI, vector databases, and LLM inference
- Built end-to-end RAG pipelines processing 5,000+ documents with sub-second retrieval latency
- Developed a multi-agent system for Travel Itinerary Agent, utilizing LangGraph for cycle management, resulting in a 35% reduction in manual oversight.
- Engineered real-time voice AI system reducing response latency by ~40%
- Designed scalable FastAPI backend handling 1,000+ requests/day
- Implemented vector search (FAISS/ChromaDB) improving retrieval accuracy by ~30%
- Worked with Snowflake/Databricks (SQL pipelines) for structured data workflows

Machine Learning Intern

Data Point Info Solution

 11/2023 - 02/2024  Hyderabad, IN

Focuses on data-driven solutions

- Developed a content-based recommendation engine using vector similarity techniques
- Performed data preprocessing, feature engineering, and similarity modeling using Pandas and NumPy
- Delivered improved recommendation quality through optimized cosine similarity ranking pipelines

RAG Engineer

Zoltan Properties

 02/2024 - 02/2025  Hyderabad, IN

Real estate technology and services

- Architected a production-grade Hybrid RAG system (Vector + BM25) for 1M+ real estate records, achieving a 92% Top-K retrieval accuracy using customized chunking strategies.
- Optimized End-to-End Latency: Reduced p99 response time from 4s to 1.2s by implementing semantic caching and asynchronous FastAPI middleware.
- Advanced Information Retrieval: Implemented Query Expansion (HyDE) and Late Interaction models (ColBERT) to handle nuanced property search queries (e.g., "quiet neighborhoods near tech hubs").
- Built automated evaluation framework using LLM-based scoring to track faithfulness, relevance, and context precision
- Production Reliability: Integrated guardrails (LlamaGuard/NeMo) to prevent PII leakage and ensure responses complied with real estate Fair Housing regulations.

Junior AI Engineer (Contract)

Concentrix YTTV Process)

 05/2025 - 12/2025  Hyderabad, IN

Customer service and enterprise solutions

- Supported AI-driven customer interaction systems, focusing on real-time response optimization and automation workflows
- Improved latency and responsiveness of AI-assisted chat systems through backend and inference flow tuning
- Contributed to scalable automation pipelines handling high-volume user interactions

SKILLS

Python	Pandas	Numpy	
PostgreSQL	WEBRTC	SQL	
Snowflake	Databricks	PyTorch	
TensorFlow	Scikit-Learn	ETL	
STT	Flask	REST	API Design
SQLAlchemy	MongoDB	Docker	
Pinecone	Linux	Milvus	Weaviate
LangChain	LangGraph	LangSmith	
LoRA/QLoRA	RAGAS	LlamaIndex	
Azure	Linux	Git	AWS

CERTIFICATION

GENERATIVE AI AND AGENTIC AI DEVELOPMENT

Boston Institute Of Analytics

EDUCATION

Intermediate College

Narayana Junior College

04/2019-03/2021  Hyderabad, IN

Bachelor of Engineering (BE)

Osmania University

 04/2021-05/2025  Hyderabad, IN